

# Why representation(s) will not go away: Crisis of concept or crisis of theory?

RENÉ JORNA and BAREND VAN HEUSDEN

## Introduction

A treatment of the notion of 'representation' requires philosophical, conceptual, and empirical arguments (Goodman 1981 [1968]; Rorty 1980; Jorna 1990). Philosophical and conceptual arguments are always involved because the choice of concepts often presupposes a philosophical position. In addition, discussions about 'representation' and, especially, the so-called 'crisis of representation' are subject to fads and fashions. However, in our view, there can never be a crisis of 'representation', just as there can be no crisis of weather or crisis of the atom. It is true that a crisis in the interpretation of representation or in the operationalization of this notion may be at issue, but such an interpretation must be clearly formulated and elaborated. We believe that there is a philosophical debate on the notion of 'representation', but we also believe that 'representations' have a steady empirical-constructivist side that will continue to exist as long as knowledge and human cognition exist.

Let us begin with a few general remarks on 'representation'. We postulate the equivalence of the concepts of representation, symbol, and sign and support the classical tenet that representation means 'aliquid stat pro aliquo'. On these premises, we will discuss the major positions in the debate on the crisis of representation. We will then go into the details of various aspects of 'representation', such as its characterization in terms of predication, various features of the notion of representation, and the relation between representation, knowledge, and human cognition. After the discussion of these analytical and conceptual aspects of our topic, the constructivist-empirical domain of representations will be dealt with in the light of our present research, the CASTOR Project, which deals with innovation and knowledge change in members of organizations. As our present CASTOR research shows, knowledge in organizations is really knowledge of human cognition and representations.

### Aspects of representation

In linguistics, philosophy, (cognitive) psychology, mathematics, and computer science, representation has meant, and still means, that something stands for something else (Rorty 1980; Eco et al. 1988). In our view, the so-called crisis of representation has resulted from attacks on this classical definition, which have not succeeded in replacing the concept of representation. Before we can go into details, let us focus on some aspects of the notion of representation.

#### *A definition of representation: Domains and relations*

The classical definition of representation is that something stands for something else: 'aliquid stat pro aliquo', as the Latin authors put it (Palmer 1978). This general statement covers at least two aspects of representation. The first is representation in the sense of a description or a formal scheme, i.e., a symbol set or a group of entities. The second is representation in the sense of mirroring or depicting representation which, in this sense, constitutes a relation. A third sense can be discerned in psychological terminology: representation as a procedure or process also called (internal) action. The relational and procedural senses are similar insofar as both conceive of representation as a (cognitive) function or a mechanism. The difference is that the materialization of representation is left open in the case of depiction while, in human information processing, representation has a cognitive realization.

According to Palmer, who has formulated several properties relevant to systems of representation, there are two separate but connected domains in a representation: namely, a represented domain and a representing domain. In every case of representation, one must specify (a) the represented domain and (b) the representing domain. Furthermore, one also has to know (c) which elements of the represented domain and (d) which elements of the representing domain are involved in the representation. Finally, one has to determine (e) the correspondences between the domains (Palmer 1978: 262).

Let us take the albeit counterintuitive example of the design and production of a new automobile in a car plant to explain Palmer's criteria. The design department will first deliver an elaborate blueprint of an automobile. This blueprint consists of a complicated set of signs with a whole range of specifications, prescriptions, and rules. On this basis, a prototype of the car will be built with the various design specifications. In Palmer's terms, the design is the represented domain, the car itself is

the representing domain, and elements of both domains constitute the representation. For example, from the represented domain, the structure and measures are important, whereas the graphic material of the blueprint does not belong to the representation. From the representing domain, the prototype, the place of the steering wheel is important to the representation, but not its precise position. Determining the correspondences between both domains is also possible, although, in so doing, one must keep in mind that the prototype is the depiction, whereas the design is the object. The represented and representing domains can be clearly determined and measured. If errors occur in the prototype (the representing domain), they can be easily traced back to misinterpretations of the represented domain. Perhaps the signs and symbols used in the designs are not clear enough, but since the elements in both domains can be well specified, errors can be corrected.

In this example, the represented domain (the design or blueprint) consists of signs of schemes, diagrams, and verbal descriptions, whereas the representing domain is the prototype, which consists of material such as wires, tin, steel, or plastic material. This is why the example is counterintuitive. It is usually assumed that representation, as a relation or function, naturally goes from the material world to the world of signs and symbols. The direction of the relation, that is to say, the specification of the function, determines the representing and the represented domains. There is no reason that the one direction is more natural than the other. For this reason, Palmer emphasizes the importance of characterizing representation as a relation. The function may be from the material to the design, but also from the design to the material.

Hence, the domains in our example are observable or, to a certain extent, objective: irrespective of the direction of the relation, diagrams and designs in the form of schemes can be read, and objects can be observed and possibly even manipulated. Difficulties arise when one of the domains is cognitive or mental. In this case, determination or observation is no longer directly possible, and indirect specifications have to be established or derived. In cognitive science, cognitive psychology, and artificial intelligence, this peculiarity of 'being cognitive' of at least one of the domains of representation has been a major topic of research (Gardner 1985; Jorna 1990).

In this context, Newell and Simon (1972) formulated their hypothesis of the physical symbol system. It states that human thinking and reasoning consists of the manipulation of (internal) symbol systems. Symbols or representations are the basic constituents of our thoughts, and manipulations are operations such as to create, copy, delete, move, and integrate symbols. The symbols or representations are treated as

functional entities, but they have a material carrier. Therefore, the attribute 'physical' is used. In human information processing systems, the material substrate is the brain. In computers it is the physical processor.

Representation is not only meant to cover the (internal) symbols in their functional roles, but also the procedures and processes within the cognitive system. Often it is difficult to unravel the sophisticated distinctions concerning representation, its domain(s), its relations, or the mathematical functions between the domains and the internal processes that take place in a cognitive system (Kosslyn 1980; Pylyshyn 1984; Jorna 1990; Jorna 1997). However, the positional variations are not relevant in the discussion about the crisis of representation. What is important is that the character of at least one domain, i.e., the cognitive or mental domain, is relevant. We will come back to this issue later.

#### *Technical and psychological properties of representation and predication*

Although often neglected, the basic claim that a representation is something that stands for something else has not remained undisputed. In technical terms, the question is whether representation always consists of the dyadic relation or a two-place predicate. Goodman (1981 [1968]) has argued that it is incorrect to conceive of a representation as a two-place predicate. A picture or a description of a unicorn, e.g., is a representation with a zero denotation since, although the picture is the unicorn, unicorns themselves do not exist. How is it possible that something represents and, at the same time, represents nothing? As Goodman (1981 [1968]: 121) argues: 'What tends to mislead us is that such locutions as "picture of" and "represents" have the appearance of mannerly two-place predicates and can sometimes be so interpreted. But "picture of Pickwick" and "represents a unicorn" are better considered to be unbreakable one-place predicates, or class-terms, like "desk" and "table"'. According to Goodman, some representations do not depict; they only categorize and, hence, only refer to themselves. By defining them as one-place predicates, Goodman wants to emphasize that a representation does not necessarily have to refer at all, i.e., a representation must not necessarily stand for anything. Whenever a representation is a one-place predicate, it is not a depiction but a classification, which means that we are talking about a property or a feature of something. In this case, it is better to speak of a *representation as* than to speak of a *representation of*. Hence, Palmer's relational aspect of representation disappears together with one of the domains, leaving only the other, and this increases the interest in the

domain as a structure in itself, since we can now only analyze a 'syntactic', not a semantic structure.

Even if representation is conceived of as a two-place predicate, it always raises the question of the entities of the domain. Let us suppose that we have a photograph of Bill Clinton. The photograph (*y*) refers to the former President of the United States, that is to say, it is a two-place representation of *x* (the president). But the photo is also a photo-representation (*y*-representation); in short, a photo is also a one-place representation. It is, for example, neither a sculptured nor a knitted representation. According to Goodman, this means that 'in general, then, an object *k* is represented as so and so by a picture *p* if and only if *p* is or contains a picture that as a whole both represents *k* and is a so-and-so-picture' (1981 [1968]: 129). Representing does not necessarily mean reflecting or mirroring (Rorty 1980). It is a symbolic relation which is variable and relative.

Now let us suppose that someone says that Clinton's photograph not only represents the former president, but, in the eyes of a conservative southerner, also a filthy liberal. In this case, we can say that the photograph *as* a pictorial depiction is a representation *of* someone (the president) *for* all people and at the same time someone else (a liberal) for a selected group of (other) people. This makes representation a three-place predicate: *y* is a representation of *x* *for* *z*.

The analytical distinction between representation *as*, representation *for*, and representation *of* has rarely been used, with the exception of Charles S. Peirce's semiotics. Representation, according to Peirce, is a three-place predicate: a sign *a* stands for, or represents, an object *x* in relation to an interpretant *b*. The sign consists of the three-place relation of a quality with an object and an interpretant. Peirce says that a sign, also called a representamen, creates something in the mind of someone, which he calls an interpretant. One has to realize that the interpretant is a logical entity, not a cognitive system. Furthermore, a sign stands for something which Peirce calls the object. Besides, the sign stands for the object in a certain respect, that is to say, it represents the object not in all its aspects, but only in relation to a certain idea which Peirce calls the quality or the ground (CP 2.308). Representation as a three-place predicate thus implies that something stands for something else. For instance, if I say that the picture of my old-timer is beautiful, I implicitly mean 'to my mind'. Hence, representing implies a (cognitive) system that 'does the work'. We will come back to this 'necessity' later on.

The determination of the various domains involved in a representation can be further elaborated in terms of the elements or entities involved. Whenever a scheme, diagram, picture, or description describes

something, the characteristics of the elements of these signs can be discussed. Not every sign or set of signs has the same semiotic potential. What makes one sign set different from another? What is the difference in terms of sign characteristics between a picture and a description of Bill Clinton?

A first step toward an answer to these questions is the recognition that not every sign set is a notation. In set theory, e.g., a set of mathematical symbols forms a notation, whereas a set of pictograms does not. According to Goodman (1981 [1968]), a notation fulfills five syntactic and semantic requirements, of which its most important function is its authoritative identification. Consider the example of a musical score: 'A score, whether or not ever used as a guide for a performance, has as a primary function the authoritative identification of a work from performance to performance' (Goodman 1981 [1968]: 128). A sign which is an element of a notational system makes it possible to decide whether we are talking about the same thing on different occasions. Signs can be categorized according to their syntactic and semantic characteristics. There are two syntactic and three semantic requirements for sign sets to form a notation, according to Goodman.

The first syntactic requirement is disjointness: the signs in a symbol set must be real copies or replicas of each other. Consider the characters of the alphabet. The character */-*, e.g., can indicate an A or an H, but for symbols in a notation, this situation is disastrous because the sign cannot be attributed. A sign which does not belong to either character violates the criterion of the disjointness of symbols. 'Two marks are character-indifferent if each is an inscription (i.e., belongs to some character) and neither one belongs to any character the other does not. ... As a result, no mark may belong to more than one character' (Goodman 1981 [1968]: 131–133). We need a symbol set which can help decide whether a scratch or a sound is a replica of one of the signs constituting the symbols in this set.

The second syntactic requirement is finite differentiation. Goodman's definition is: 'For every two characters *K* and *K'* and every mark *m* that does not actually belong to both, determination either that *m* does not belong to *K* or that *m* does not belong to *K'* is theoretically possible' (1981 [1968]: 135–136). Examples of sign sets with finite differentiation are the alphabet, the Arabic number system, and the binary code. Both syntactic requirements ensure that signs can be substituted for one another (disjointness) and that one can decide whether two characters belong to the same sign (differentiation). Where syntactic requirements are concerned, spoken and written language, the Arabic number system, and musical notes are notational schemes.

Besides the syntactic, Goodman postulates three semantic requirements. The first is that a sign system has to be unambiguous: no element of the sign set may refer to different 'things' (which the nominalist Goodman calls 'compliance class'), the reference relation is constant, and a sign has just one extension. Hence, 'any ambiguous inscription must be excluded since it will give conflicting decisions concerning whether some object complies with it' (Goodman 1981 [1968]: 148–149).

The other two semantic requirements are disjointness and finite differentiation. Semantic disjointness means that a sign set must not be redundant. There is redundancy if two signs refer to one and the same compliance class so that there is not just one class, but an inclusion of one class within another. Unambiguity and semantic disjointness are, in a way, complementary because the former guarantees that one sign belongs to one compliance class, while the latter assigns one sign to one compliance class only.

A notational system also requires finite differentiation: 'For every two characters  $K$  and  $K'$  such that their compliance-classes are not identical, and every object  $h$  that does not comply with both, determination either that  $h$  does not comply with  $K$  or that  $h$  does not comply with  $K'$  is theoretically possible' (Goodman 1981 [1968]: 152). A sign set that is not finitely differentiated is semantically dense. Goodman formulates this third requirement because whatever a sign set denotes must at least theoretically be distinguishable.

If a sign set is syntactically and semantically disjoint, finitely differentiated, and, furthermore, semantically unambiguous, the sign set may be called a notational system or a notation. In all other cases, a sign set is not a notation. Musical notation, the ZIP code, or Laban notation for dance and movement can each be characterized as notations, but neither the alphabet nor various kinds of pictorial languages are notations in the strict sense, since they lack one or more of these requirements.

Goodman also mentions other features of signs that allow the comparison of symbol sets in terms of their communicative efficacy. A sign system may consist of a large or a small set of signs. This difference is important in learning a sign system. One only has to compare the Roman with the Chinese alphabet in order to see the importance of the magnitude of a set. Also, the ease with which signs can be made and manipulated, their readability, their graphical suggestiveness, and their mnemonic efficacy play a large but often neglected role in research into sign systems. The growing importance of communication with screens and represented realities in which command languages, diagrams, graphical forms, and other sign systems are used testify to these nontechnical or psychological features of sign sets (Jorna 1988). The features can be

Table 1. *The sign sets of icons and verbal language with respect to their features*

| Features                         | Icons/pictorial representation        | Language/word representation |
|----------------------------------|---------------------------------------|------------------------------|
| <i>Technical features</i>        |                                       |                              |
| syntactic disjointness           | no                                    | yes                          |
| syntactic finite differentiation | no                                    | yes                          |
| semantic unambiguity             | no                                    | no                           |
| semantic disjointness            | no                                    | no                           |
| semantic finite differentiation  | no                                    | no                           |
| <i>Psychological features</i>    |                                       |                              |
| size of sign set                 | large/small (context dependent)       | small (characters)           |
| clarity                          | low (needs interpretation)            | high                         |
| graphic suggestiveness           | high                                  | low                          |
| mnemonic efficacy                | high                                  | high                         |
| performability                   | difficult (to make);<br>easy (to use) | easy (to make and to use)    |

treated in two ways. The first is that they are descriptive dimensions that can be discerned neutrally. However, if one wants to look at the prerequisites for unambiguous communication, the features become requirements for sign sets.

As a first approximation not based on empirical analysis, Table 1 gives a rough survey of the technical/semiotic and psychological aspects of two signs sets, the pictorial signs of the Win98 interfaces and the language signs of verbal communication (characters, words, and sentences). The values for the technical/semiotic and psychological features in the matrix show important differences. Neither the pictorial nor the verbal signs are notations. They cannot be used for error-free communication. As is well-known, computers do not use these sign sets. They work by protocols, which are notations. The psychological features are related to the sign users and their interpretations. If the set of icons is relatively small, e.g., in the various bars of word processors, efficacy and graphic suggestiveness are high. We will not elaborate the details here, but they give many opportunities for empirical semiotic research.

### *Kinds and types of representation*

For the moment, we consider representation to mean that something stands for something else, being a two-place predicate. In everyday life,



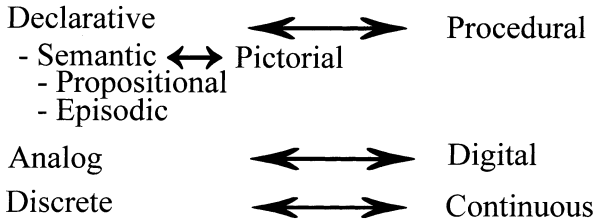


Figure 1. *The interrelation of various forms of representation*

we often forget what the sign refers to, since we use representations as signs to make things less complicated. Often a representation is a descriptive scheme, which can be characterized in terms of the entities used. Hence, we speak of semantic and pictorial representations, but many others can be found (Simon 1978; Kosslyn 1980). The interrelation of several forms of representation is depicted in Figure 1, in which forms of representation are structured with respect to their basic constituents and their function. The main dichotomies are declarative vs. procedural, analog vs. digital, and discrete vs. continuous. Semantic, pictorial, propositional, and episodic representations are all declarative. Semantic representations presuppose that meanings of mental elements consist of sets of verbal associations structured in the form of networks. Several semantic systems have been formulated for this purpose. Episodic representations (Tulving 1983) consist of events or episodes structured along a temporal dimension, the structure of events having an internal hierarchy. Propositional representations consist of statements with a truth-value to which the rules of a logical calculus are applicable (Fodor 1975). Pictorial representations are rather similar to images in visual perception (Kosslyn 1980; Johnson-Laird 1983). They can also be considered declarative. However, in contrast to propositional representations, they have no truth-value. The main characteristic of declarative representations is that they are explicit, i.e., their knowledge structure consists of symbols which are directly accessible. This is not so with procedural representations, which are related to operations. Procedures are not directly accessible. In fact, the distinction between declarative and procedural representations takes place on a different level than the one of the distinction between semantic and pictorial representation.

In Figure 1, there are two further distinctions: analog vs. digital and discrete vs. continuous representations. The analog vs. digital distinction is based on the criterion of similarity, whereas the discrete vs. continuous distinction focuses on the continuity of physical reality — whether taken

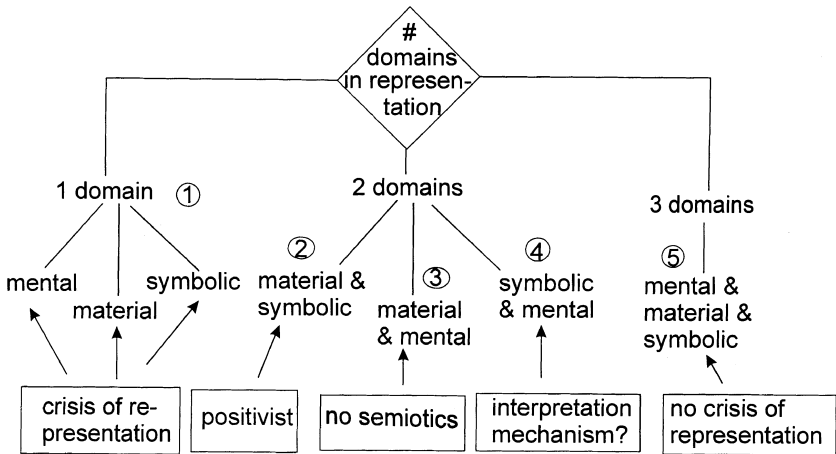


Figure 2. *Domains of representation and the so-called crisis of representation*

over into the represented world or not. Finally, there is much overlap between analog, continuous, and pictorial representations on the one hand and digital, discrete, and propositional representations on the other. More fine-grained distinctions can be made, but for the moment, it may suffice that the constituents of the representing domain can be classified. A different categorization in relation to the development and conversion of knowledge in organizations will be discussed later, in the last section of this paper.

### **The crisis of representation: Positions and claims**

Statements about representation have philosophical underpinnings and consequences which we cannot consider exhaustively here. However, we will focus on two modes of theoretical branching in the study of the domains of representation: the representing and the represented domains (see discussion in our earlier section of this paper, 'A definition of representations: Domains and relations'). The first mode occurs when we consider that one, two, or three domains may be involved (Figure 2). The second mode concerns the further branching of these domains into mental, material, or symbolic representations. In principle, these two kinds of branching are independent of each other. When only one domain is considered, neither mental, nor material, nor symbolic representations can be discerned. When two domains are involved, they branch into symbolic plus mental, material plus mental, or material plus symbolic.

With three domains, mental, material, and symbolic are separately present (Figure 2).

Philosophically, the problem of representation cannot be resolved. Conceptual analysis may help to clarify positions, but only empirical research can bring a solution. However, even empirical research can be criticized because of a philosophical bias in its methodology. In our view, the so-called crisis of representation results from the incompatibility of divergent philosophical positions. Figure 2 distinguishes three main positions according to the numbers of domains involved. To the left of the diagram (see [1]), we see the ‘one-domain position’ in the theory of representation. It claims that there is only one (large) domain of representation, the universe or ‘reality’, which consists of one kind or type of entities with subdivisions into material, mental, or symbolic representations. This position raises various problems, especially in the symbolic and mental subdomains. First, there is the question of whether we still have representation at all, since it might be argued that we have only presentation. Secondly, the so-called classical distinction between subject and object has vanished, according to this position. Thirdly, this position ignores reference, which implies that truth determination becomes almost trivial. The sole criterion of this approach to representation is consistency within the domain. According to this position, technical and psychological features of the elements of representation are relevant when the domain is symbolic or sign-oriented. We can only speculate on what it means to say that the domain is either material — representations in whatever meaning having disappeared — or mental. The idea of representation within the mental domain alone leads to solipsism. With regards to Goodman (see our previous section, ‘Technical and psychological properties of representation and predication’), representation in the symbolic domain alone exists only as classification (representation *as*). Reference is only internal, from sign to sign.

According to the first two-domain position shown at the center of Figure 2, representation involves two separate domains, the material and the symbolic one, each with their own specific characteristics. ‘Material’ refers to all objects in the world, whether natural, constructed, artificial, or otherwise. ‘Symbolic’ refers to all sign types from icons and diagrams to words, characters, and mathematical symbols. This position has several consequences. First, a representation is a two-place predicate, relating the material world to the sign. It is a representation *of*. Secondly, there is a domain of reference. The usual route of reference is from the sign to what it refers to, but the reverse route, from the object to the sign, is still possible, as in our previous counterintuitive example of the design of a car and its prototype (see our earlier section, ‘A definition of representation: Domains

and relations'). Thirdly, we can work with the classical correspondence theory of truth. Furthermore, there is still the issue whether the relation is unilateral, from sign to object or object to sign, or bilateral, i.e., both ways at the same time.

The third, fourth, and fifth positions shown in Figure 2 also acknowledge two different domains, but at least one of them is a mental or cognitive one. The underlying theory of representation is thus called the mental-domain position. It differs from the second position, where mental or cognitive considerations were deliberately left out. We can distinguish at least three forms of this third position: a combination of (a) a mental with a material (see [3]), (b) a mental with a symbolic (see [4]), and (c) a material with a mental and a symbolic domain (see [5]). Some might want to argue that the mental and symbolic domains cannot be separated, but then the discussion is about whether signs by definition need interpretation. (We will argue later in favor of this last point of view.) There are three consequences of this position. First, there is a subject-object distinction. Secondly, we have reference, although this is more problematic than according to the second position. We have also a correspondence theory of truth. Thirdly, interpretation depends on a cognitive system, which is not unproblematic since it raises issues of cognitive architecture, mental processes, and mental representations.

Several characterizations have been given to the five positions outlined above. The one-domain position has been called nominalist, idealist, materialist, solipsist, or, more recently, deconstructivist. The first two-domain position has been called positivist, empiricist, rationalist, or realist. The difference between the other two- and three-domain positions reflects the importance given to the cognitive system. The third, fourth, and fifth — mental domain positions — are characteristic of the cognitivist, the connectionist, and also the situated-action models of the mind. Not wishing to take a stand in these matters for now, we will confine ourselves to saying that we favor this mental-domain position.

Let us turn to the question of whether there is a crisis of representation. The sophisticated answer is that it depends on the position. From the semiotic perspective, a crisis of representation can only be ascertained in the one-domain position where texts refer to texts which refer to other texts, resulting in the so-called unending semiosis. Various examples of this 'crisis' can be found in recent deconstructivist and poststructuralist literature based on Derrida (1967). The notion of representation is deconstructed in terms of signs, other signs, concepts, and further signs. It has been said that the notion of 'being about' as a part of representation should be replaced by the notion of 'difference'. However, without independent criteria, this approach leads to going around in circles.

Ultimately, it means that a literary text or a piece of art only refers to itself. In this sense, there is a crisis of representation. No such crisis of representation can be discerned with the other four positions, although there is the problem of interpretation with the two-domain position.

As long as we are in a philosophical debate, there can be no conclusive answers. It is true that in the field of arts and literature, the notion of representation may be in crisis, since 'representation' is treated in a perverse way without taking into account science, cognition, and knowledge. If one looks at the (scientific) fields dealing or working with the notion of representation, e.g., semiotics, cognitive science, mathematics, or computer science, there is no crisis. Here, the concept of representation is solidly founded in epistemological positions and in the way we perceive our cognitive system: leaving cognition out would dehumanize semiotics. Humans always use and process representations. Therefore, the crisis is not about the phenomenon, but about the theory of representation. The crisis is a crisis of the representation of representation. Hence, crisis means that we are misled by theory. As far as there are problems with representation, they concern the operationalization of representation. This crisis may be handled by making an empirical turn toward a semiotics of cognition (and culture). That is, we still have the phenomenon, and we have to operationalize it in a better way. Representation is about knowledge and, in one way or other, it requires a cognitive system. Therefore, we believe that a discussion of representation is mainly a discussion of knowledge.

### **Representation and knowledge in organizations: The CASTOR Project**

We have argued that the crisis of representation is a crisis of a certain conceptualization of representation and thus a crisis of the interpretation of the *concept* of representation. We have also shown that it is possible to continue working with the concept of representation according to other interpretations. In particular, approaches that postulate an interpretation mechanism in the form of a cognitive system are relevant to the theory of representation. Thinking, reasoning, decision making, and problem solving are activities of human cognition. The conceptual framework of the theory of cognition consists of cognitive architecture, representations, and operations on these representations (Posner 1989; Newell 1990; Anderson 1995). If a human subject is asked to determine whether the back seat of his car has an ashtray, he or she constructs a mental image of the car and searches this mental image for an answer. A comparable mental process takes place if a planner in problem-solving

process has to determine whether certain compound products can be produced by certain machines in a flow-job structure. In both cases, representations are empirical in the sense that people use them. This means that representation is not only the subject of conceptual analysis but that it is also the subject of empirical research and, consequently, of operationalization.

We take for granted that people in organizations are cognitive systems. We also take for granted that they have knowledge, use knowledge, and change their knowledge. If being a cognitive system means that one has and uses knowledge, it follows that one uses representations. Performing intelligent tasks in organizations requires the use of knowledge and representations. At the level of individual task performance, this has led in the recent past to research into cognition, artificial intelligence, decision support, and knowledge systems (Klein and Methlie 1995). At the organizational level, i.e., at the level of the coordination of tasks, processes, and staff, this has very recently led to themes such as organizational learning, knowledge creation, and knowledge management (Nonaka and Takeuchi 1995). The management of knowledge and learning has become an issue in management studies and economics. As shown elsewhere, management of knowledge means managing intelligent people performing complex tasks, individually and in groups, structured in processes and functions (Jorna and van Heusden 1998; van Heusden and Jorna 2000). If people intelligently perform tasks with or without the help of 'intelligent' software, they use their cognitive apparatus. Therefore, management of knowledge is management of representations. This implies that representations are not only a conceptual issue, but that we can also work with the measurement of representations.

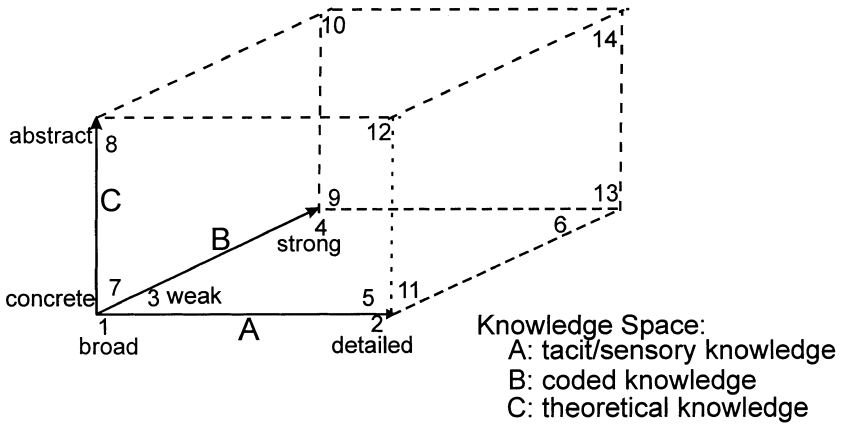
People participating in an organization have shared representations of their interactive behavior. These representations constitute the foundation of the organization. Our knowledge of reality is a semiotic, that is, a representational construction. As a (cognitive) mental activity, cognitive science has studied this representational behavior. As a representational activity, however, the cognitive mental behavior is the object of semiotics. Thus, cognitive science and semiotics meet in an effort to understand man, the animal *symbolicum*, and human culture (Jorna and van Heusden 1998). In our CASTOR Project (CASTOR stands for 'Knowledge Conversion And Semiotic Theories Of Representation'), we have chosen to follow both the semiotic and the cognitive route in an investigation of organizations and, in particular, of the management of organizational knowledge.

According to our approach, organizations in the broadest sense of the term are systems of a specific kind of interactive behavior, i.e., multi-actor

systems. These systems have a certain complexity, which is assumed to be a function of the number and types of behavior involved. Considered as a semiotic system (or set of semiotic systems), an organization results from processes of semiosis in individual actors in a context that is both physical and social. As a semiotic reality, an organization is an aspect of the knowledge available to and shared by a number of individual actors. These actors have knowledge and learn. The result is a complex representational pattern in the organization. Furthermore, the behavior of organizations is always adaptive. Such adaptive behavior is both stable and innovative; that is, it involves both the continuous exploitation of given knowledge and the exploration of new knowledge (Nooteboom 1996). Moreover, adaptive behavior in organizations can relate to the knowledge within organizations as well as to the organization as knowledge. Taking a semiotic viewpoint as our point of departure, we consider organizations to be semiotic constructs and processes, i.e., representations used by actors to organize the interaction with their social and physical environment. An organization is always an organization for actor(s) in a particular historical context, and the study of organizational behavior comes down to the study of cognitive semiotic behavior of groups of individual actors.

All semiotic activity comes down to connecting knowledge (stored in memory) with information. Information always relates to memory, but the perception of information as information presupposes a difference between memory and information. Owing to this difference, the knowledge stored in memory can become a sign. We can discern three types of semiotic knowledge stored in memory: tacit or sensory knowledge, coded knowledge, and theoretical knowledge. This typology of knowledge refers to the number of semiotic dimensions involved in representation. The semiotic space depicted in Figure 3 comprises three axes with two extremes each; (A) broad–detailed, (B) weak–strong, and (C) concrete–abstract. Points 1–14 refer to the boundaries of the line (one axis), the plane (two axes), and the space (three axes). It is important that the cube representing the semiotic space is a metaphor. The axes themselves are not orthogonal.

The starting point of a semiotic process is always the perception of difference(s) in our sensory representation. The first semiotic move is to recognize the situation in terms of a situation (or state of affairs) one already knows. Clearly, the greater the perceptual problem is, the more difficult it becomes to find an analogue. The perceptual, one-dimensional sign is the most primitive form of semiotic phenomena. The single dimension that allows us to speak of it as a semiotic phenomenon is the difference that separates memory from actuality in the process of

Figure 3. *The semiotic space*

representation. It is at this point that the semiotic gradually emerges from the nonsemiotic, where memory and actuality are still one (van Heusden 1999). Thus, memory turns into a sign representing an actual state of affairs on the basis of similarities. The semiotic process consists in relating the memory of an event with the event itself, which allows, of course, the recognition of the event in terms of another, earlier event. The process in which the two events are related is necessarily a process of transformation: reality and sign are still of the same concrete nature, and the actual appears as a transformation of the remembered. It should be clear that, by definition, every representation that is semiotic requires two domains.

At this stage, there is not yet any coding, i.e., the phenomena are not yet related to a larger set of similar events. There is only the association of a given state of affairs with a remembered state of affairs. Tacit knowledge belongs to this type. It is basically concrete and therefore also dependent on the presence of a context. It cannot be coded, it is about concrete experiences, and it can be shared only with those who are co-present. Diffusion, i.e., the communication and sharing of knowledge between actors, is a matter of imitation. Quantification of this type of representation or knowledge is possible through the measurement of details: the more detailed a representation is, the higher the level of representational knowledge. A professional will be able to represent more sensory details when looking at a certain activity than a dilettante. The dilettante, in turn, will probably perceive more than the novice, who is new to the field.



The semiotic phenomenon becomes two-dimensional when the code intervenes as a new relation between the two events in the process of representation. With this second dimension of semiosis, which is the dimension of substitution, the arbitrary and the conventional make their appearance. Substitution presupposes social convention as its warrant. Codes emerge with this two-dimensional sign, since a code is a convention establishing a relation of substitution. The two-dimensional sign both relies on and facilitates communication. It is important to realize that the second dimension presupposes the first. To be substituted by an arbitrary sign, a situation or event must first come to mind. To do so, it must differ from the mere memory of the event. Thus, the two-dimensional sign both presupposes and is based upon the one-dimensional sign.

In this second phase, events are not just recognized. They are also categorized, since 'things' are given names. Without going into detail, we can say that categorization is conventional and rests upon the grouping of features. Externalization, i.e., the conversion from tacit to explicit knowledge, requires coding. In its use of images and metaphors, this coded knowledge comes closest to the noncoded concrete knowledge of the first phase. The coded sign that remains most tied to the concrete event is probably the conventional gesture, e.g., the gesture of pointing. Taxonomies are a good example of coded knowledge. The two-dimensional sign has a context, but this context is no longer the one of a concrete state of affairs, having extended to comprise the group of actors who know and use the same code. Coded knowledge can be quantified by taking into account the number of its elements (words, numbers, pictorial elements), its combination rules (or grammar), and the degree of ambiguity allowed by this code (cf. our earlier section, 'Technical and psychological properties of representation and predication'). The system of traffic signals, e.g., consists of a small number of elements, has a simple grammar, and allows for little ambiguity. At the lowest level of codification, codes tend to dissipate into concrete, that is, one-dimensional sensory knowledge (Goodman 1981 [1968]; Jorna 1988).

The three-dimensional sign emerges when a third dimension is added to the ones of sensory difference and conventional substitution. This third dimension is the one of logical or structural relation. Whereas the one-dimensional sign relies on a process of transformation and the two-dimensional sign on a process of substitution, the three-dimensional sign relies on the analysis of relations or structures. This third dimension arises when coded signs relate to the events represented, not on the basis of a convention, but on the basis of formal or structural qualities (such as their number). The result is a new dimension of theoretical knowledge. Once

this third dimension is added to the other two — and again, addition is cumulative: no logic without perception and codes — a hidden reality of meaning emerges as a third factor in the semiotic process. Concepts are added that no longer refer to classes of objects, but to formal relations such as ‘number’ and ‘cause’. In the third phase, knowledge is knowledge about the necessary relations between events or categories of events. Scientific and mathematical knowledge belongs here. Knowledge is now the result of scientific inquiry — empirical as well as theoretical, inductive, and deductive. It should be clear that theoretical knowledge makes diffusion even easier than coded knowledge does. Knowledge is not, in principle, conventional. It is universal.

Theoretical knowledge can be quantified by measuring the complexity of the algorithms involved in problem solving. Algorithms can be evaluated on the basis of various criteria. Problems consist of a description of the parameters involved and an explanation of the characteristics the solution must have. As yet, however, we do not know how to do this. What we can do is to bring together a typology of organizations and a theory of representation, conceived of as a basic theory of human action. The strategy of representation is reflected in the structure and ‘behavior’ (metaphorically speaking) of organizations.

The various types of sensory, coded, and theoretical knowledge are distributed spatially and temporally in organizations, and hence, not all of them are always present. The distribution of the types depends on the individuals working in an organization. It can be argued that bureaucratic organizations have more members with coded knowledge, whereas the members of newly emerging organizations work more with tacit knowledge. Individual types of knowledge are also subject to change over time.

### **Empirical studies and conclusion**

As far as knowledge management is concerned, we consider it an empirical science which involves representation and cognition. An empirical science consists of theories and hypotheses. It also uses observations and instruments, and it produces data.

In order to describe and analyze the knowledge involved in change and innovation, we developed and adjusted worksheets based on the CommonKADS Methodology (Schreiber et al. 2000) from which we can assess three kinds of information.

First, we elaborate an overall characterization of the organization with information about the structure and sequence of processes from a bird’s-eye view. At a deeper level, the tasks, sub-tasks, actors, relations between

actors, and communication structures are also analyzed. This first part of the analysis is similar to what is often done in business re-engineering and consultancy projects. Although the information is partly provided by members of the organization, the view of the organization is the one of an outsider. The important outcome of this first assessment phase is the mapping of actors and tasks.

Secondly, a simple questionnaire is sent out to all actors involved in the tasks studied in the organization. This questionnaire is used to describe knowledge of contexts, task-related knowledge, and communication processes, and to provide us with multiple internal views on the organization and on task knowledge, which are then compared. It should be kept in mind that an organization only exists as far as it is present in the representations of actors.

Thirdly, a structured questionnaire in interview format is filled in by a subset of the actors from the second phase. Special care is taken to select actors whose knowledge will radically change because of new products, services, or work practices. The data of the third phase consist of the determination of types and forms of knowledge relevant to the actors, such as the proportions and ratios of the percentages of tacit, coded, and theoretical knowledge that actors assume to be relevant for their task execution. It may turn out that those employees who share the same function, the same background, and the same number of years of experience differ in the kinds of knowledge used to fulfill their tasks.

Finally, the second and third phases are repeated several times. This leads to what we, with a metaphor, call 'knowledge snapshots'. If more snapshots are taken, it is possible to discover distortions and shifts in types and forms of knowledge between the shots. This is the operationalization of what we have called 'knowledge conversion'. Because of the piles of snapshots, learning and knowledge management are no longer mere abstract words, but can be related to (the conversions of) the types and forms of knowledge with specific actors.

The empirical research is conducted in two environments. The first is the one of the Royal Netherlands Army (van der Voort 2000). Our aim is to explain the differences in the innovative information management between its 'green' and its 'white' staff. The 'greens' are those who are responsible for information management within the Dutch-German Army Corps, which is the operational part of the army, deployed in war or peacekeeping operations. The 'whites' are the personnel of the 'National Commando', who are responsible for the information management and support of the military operational tasks of the Corps personnel. Although there is no a priori reason why different parts of the army should have a different rate of adopting innovations, in practice,

they clearly do. With our semiotic model, we hope to give advice concerning possible improvements. The second research field is concerned with the conversion of knowledge as a result of new software in planning and scheduling support. Take, e.g., the case of a hospital ward that is accustomed to manually scheduling the personnel of a unit. When the situation changes from manual to computer-assisted scheduling, the scheduler, who is now using planning support software, realizes that in the past he had been working with more knowledge than he was aware of.

By using the semiotic model of the three types of knowledge and the empirical domains of innovation in the Army and in software implementation, we have bypassed the so-called crisis of representation. We have shown that the crisis is only a crisis in one interpretation of representation, namely, the interpretation of representation as a one-place predicate. We have also argued that representation requires cognition in one form or another, which makes the theory of representation also an empirical study, subject to operationalization. Representations are empirical phenomena as long as we accept cognition. Our domains of investigation are those of innovation in organizations where intelligent people perform tasks. In the execution of these tasks, whether with intelligent computer support or not, various types of knowledge and representations are being used. This does not prevent the conceptual analysis of these representations. On the contrary, it requires the combined conceptual and empirical analysis and creates the opportunity of giving semiotics an empirical direction. A different direction of research would be to deny the relevance of cognition as a mechanism for the interpretation of representations, but we are convinced that this direction is not the one that should be taken. Rejecting the assumption of intelligence and cognition in humans as information processing systems is theoretically possible, but it raises the question: who sees in the crisis of representation a crisis and who sees in it representation?

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René Jorna (b. 1953) is Professor of Cognitive Science at the University of Groningen <R.J.J.M.Jorna@bdk.rug.nl>. His research interests include planning, cognition, and multi-actor theory. His major publications include *Signs, Search and Communication: Semiotic Aspects of Artificial Intelligence* (ed. with B. van Heusden and R. Posner, 1993), 'The SEC-system: Reuse support for scheduling system development' (with W. van Wezel, 1999), and 'Paradoxes in planning' (with W. van Wezel, 2001).

Barend van Heusden (b. 1957) is Professor of Comparative Literature at the University of Groningen. His research interests include semiotics, literary theory, and multimedia. His major publications include *Algemene literatuurwetenschap: Een theoretische inleiding* (with E. C. S. Jongeneel, 1993), *Why Literature? An Inquiry into the Nature of Literary Semiosis* (1994), and 'Semiotics of the user interface' (with René Jorna, 1996).